Exercise 3.5

Miscellaneous Practice Problems

1. Subtract: $-2(xy)^2(y^3+7x^2y+5)$ from $5y^2(x^2y^3-2x^4y+10x^2)$
2. Multiply $\left(4x^2+9\right)$ and $\left(3x-2\right)$
3. Find the simple interest on Rs. $5a^2b^2$ for $4ab$ years at $7b\%$ per annum.
4. The cost of a note book is Rs. $10ab$. If Babu has Rs. $\left(5a^2b+20ab^2+40ab\right)$. Then how many note books can he buy?
5. Factorise : $\left(7y^2-19y-6\right)$

Challenging problems

6. A contractor uses the expression $4x^2+11x+6$ to determine the amount of wire to order when wiring a house. If the expression comes from multiplying the number of rooms times the number of outlets and he knows the number of rooms to be $\left(x+2\right)$, find the number of outlets in terms of 'x'. [Hint : factorise $4x^2+11x+6$]
7. A mason uses the expression x^2+6x+8 to represent the area of the floor of a room. If the decides that the length of the room will be represented by $\left(x+4\right)$, what will the width of the room be in terms of x?
8. Find the missing term: $y^2+(\ldots)x+56=\left(y+7\right)\left(y+\ldots\right)$
9. Factorise : $16p^4-1$
10. Factorise : $3x^3 - 45x^2y + 225xy^2 - 375y^3$

3.8 Linear Equation in One Variable

3.8.1 Introduction

We shall recall some earlier ideas in algebra.

What is the formula to find the perimeter of a rectangle? If we denote the length by l and breadth by b, the perimeter P is given as $P = 2(l + b)$. In this formula, 2 is a fixed number whereas the literal numbers P, l and b are not fixed because they depend upon the size of the rectangle and hence P, l and b are **variables.** For rectangles of different sizes, their values go on changing. 2 is a **constant** (which does not change whatever may be the size of the rectangle).

An algebraic **expression** is a mathematical phrase having one or more algebraic terms including variables, constants and operating symbols (such as plus and minus signs).

Example: $4x^2 + 5x + 7xy + 100$ is an algebraic expression; note that the first term $4x^2$ consists of constant 4 and variable x^2. What is the constant in the term $7xy$? Is there a variable in the last term of the expression?

The 'number parts' of the terms with variables are **coefficients**. In $4x^2 + 5x + 7xy + 100$, the coefficient of the first term is 4. What is the coefficient of the second term? It is 5. The coefficient of the xy term is 7.

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